KU LEUVEN

Data Visualisation Project Report

[redacted]

Part 1. Metadata

· Version: Implementation (10/05/2024)

· Students:

[redacted]

· Dataset: D.E.A.D

Part 2. Project description

Dumbledilp's Equipment & Adventuring Distributions Inc. (D.E.A.D) is seeking assistance in gaining deeper insights into their product sales, regional distribution, delivery times, and overall market trends. Our task involves creating informative visuals based on data they accumulated over the past five years; empowering them to refine their marketing strategies.

The dataset consists of four Excel files, namely customers, orders, products and regions.

- Customers: There are 2498 customers in total. Every customer has a unique account code which can be a private or professional buyers (key account), a key account manager, an account name and an account manager.
- Orders: The dataset includes approximately 280,000 orders. Each order has a unique orderID and comes with an order and delivery date. We get information about the territory, products, quantities (of each product) and prices of the orders (CartPriceInCP, Cartprice (GP, SP, CP), ProductPricesInCP).
- Products: A list of the 2101 products D.E.A.D offers, categorized into 3 business units, 8 types and 42 subtypes. Brand names and managers are also provided for each product.
- Regions: Regions are hierarchically categorized, split up by area (5), region (15), nation (57) and territory (212), with a corresponding manager for each area and region.

From our initial analyses of the data, a few interesting and notable observations were made:

- The main response variable of interest was revenue (calculating by summing the order cart prices, measured in Copper Pieces). However, other response variables of interest were noted too, like average order price, delivery times and number of orders.
- The value of orders in Copper Pieces was pareto distributed, which meant that some orders were much more valuable than average.
- The prices of a product could greatly differ depending on the order (a product may cost 20 copper pieces in one order but 150 copper pieces in another order, for example).
- There was also notable variability in the delivery times across the orders.
- There were several different ways we would like to analyze the response variables; by location (area, region nation and territory), by product (type and subtype) and by manager (area and regional).
- These categories were not balanced, as the number of customers, number of orders or number of products in each category could differ greatly. This would need to be taken into account.

After carefully analyzing the dataset and taking these observations into account, we came up with three specific questions to furtherly guide our analysis:

- Which product categories are excelling and which ones generate relatively low revenue and could potentially be omitted?
- What explains the product prices fluctuating so much? Is it determined by area, year or something else?
- What were the main factors that determined delivery time? Did it differ a lot across different locations?
- Where are our most valuable orders coming from?

Part 3. Visual design

A. Diverge

After reviewing the data and considering the additional information (questions of the client), each of us created individual sketches. A week later we gathered together and shared our respective sketches. We found a wide variety of different kinds of visual representations, with each person exploring slightly different themes. We also found that the sketches ranged in complexity, with some conveying very simple pieces of information in simple ways, and others portraying lots of different factors in one visualization. The large variety in visualization types left us with lots of ideas to explore further, to combine different sketches and to alter as needed. After explaining each sketch to one another, we left the meeting with new ideas to explore. We each revised some sketches independently, created additional ones and uploaded them to Miro (see the appendix below for an overview of all the sketches). As mentioned before, there was large variation in graph complexity, with some graphs being

very simple and portraying single pieces of information, and others being very complicated and trying to incorporate as much information as possible. We found that the most informative and visually digestible graphs lay somewhere in the middle, with enough information portrayed so that unique insights could be extracted, but not so much that it resulted in information overload.

B. Emerge

Next we met online and used Miro to cluster our sketches. We clustered the graphs in two different ways, either based on what information was being shown in the graph or the format in which the information was being portrayed.

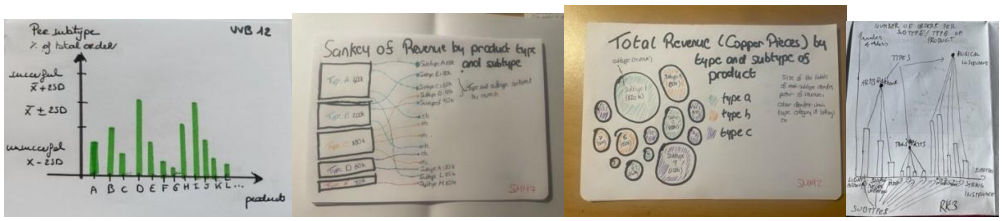
From this clustering, three main clusters emerged:

1. Analyzing revenue by type and subtype of product
2. Circular visualizations of scaled features across regions (revenue, number of orders, delivery time, kind of buyers, etc.)
3. Representations of the data by location through maps

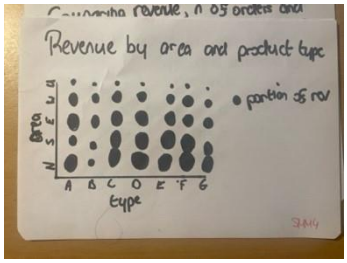
C. Converge

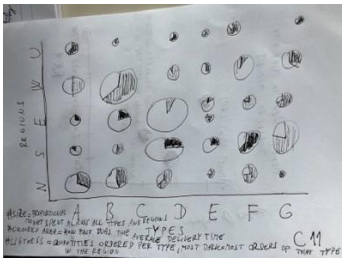
We furtherly focused on these three clusters and compared the visualizations within and between them. From doing this, we came up with one single visualization for each cluster that we felt gave the most valuable insight into the data, while adhering to good design principles. A description of the process from start to finish is given below for each cluster.

Cluster 1: Analyses by Type & Subtype



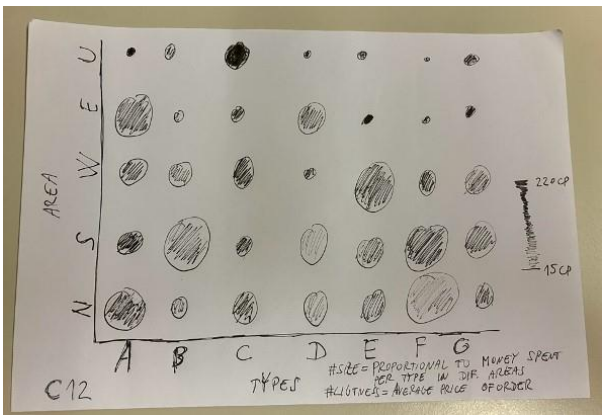
Analyzing revenue by type and subtype of product was something we initially deemed very insightful to do. This could help D.E.A.D to understand what lines of products were most important for their business and where more emphasis should be placed. To visualize this, we initially had many different types of visualizations. VVB 11, VVB12 and RK3 implemented bar charts and line plots. SMM7 used a Sankey diagram to show the order of revenue of each type and subtype of product. SMM2 used a bubble diagram, where size of each circle corresponds to amount of revenue. SMM4 used a similar idea but placed the bubbles (or circles) on a plot with the x axis as type and the y axis as area.



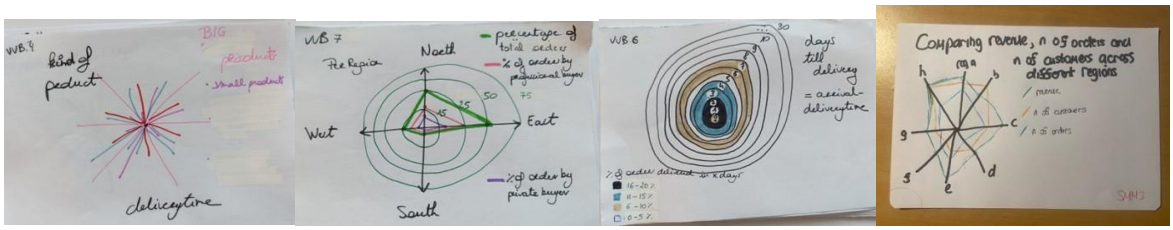


We ended up proceeding with the idea of SMM4, as it allowed us to analyze types of products across different areas, something we thought may be insightful as it would allow us to analyze trends in location and product ranges in one plot. However, we still felt more information could be conveyed in this visualization. C11 changed the circles from SMM4 into clocks, with shaded regions of the clocks representing delivery times for the area and product type.

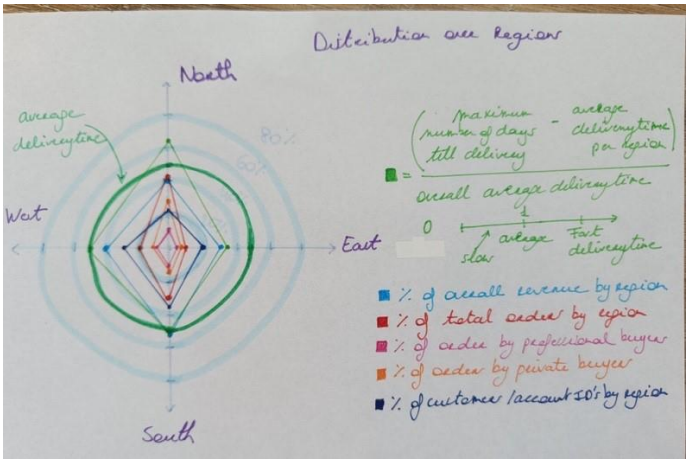
Though this made the graph more informative, we found it then became too visually cluttered, and that something between SMM4 and C11 in complexity needed to be formulated. We also noted that we still had no visualizations that gave insight into what was causing product prices to fluctuate, something we previously wanted to focus on. Thus, instead of focusing on delivery times, we could look at average order price for specific product types and see if this differed across areas (thus checking if fluctuations in product prices was a result of the area they were sold in). The finalized graph is very similar to C11 and SMM4, except now each individual circle has a different shade, with a darker shade representing a more expensive average price for a product type in a certain area. This helps to understand how much the prices between types differ and by which criterion (type or area). This can give insights about fluctuating prices of products and performance of products and also give us an idea of money streams in the company.



Cluster 2: Circular Visualizations



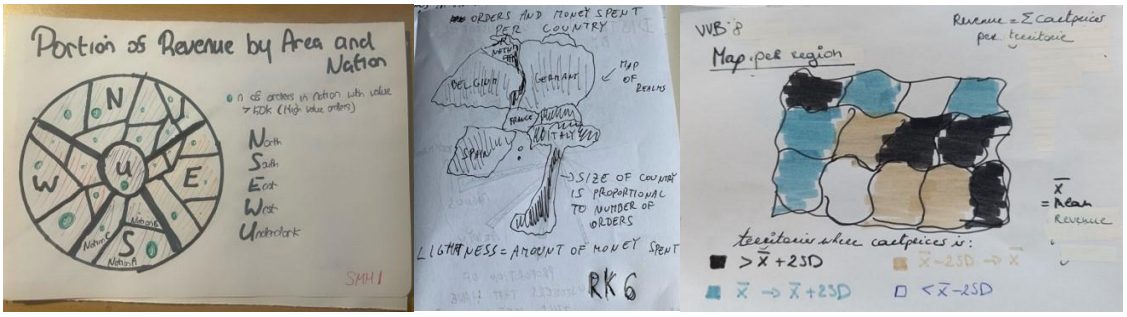
We had several sketches aiming to visualize information by means of a circular graph. We decided it would be very informative and clear to incorporate all of this information into a single graph. SMM3 focused on the distribution of revenue, number of orders and customers per region, while VVB4 and VVB6 illustrated differences in delivery times. After viewing SMM3, we proposed merging VVB2 and VVB6 by incorporating axes representing the different regions to VVB6. Building further upon this idea, we aimed to integrate all features of SMM3 and VVB2 into a single sketch. Expanding further we also wanted to incorporate insights from VVB14 and VVB15, which looked at differences between private and professional buyers. Upon looking at the data, we found no customerIDs without corresponding orderIDS. Therefore, it didn't seem necessary to investigate lost customers and determine whether they were predominantly private or professional buyers. So we left out the comparison between customers before and after the 5 year mark. However, we still incorporated the proportion of private and professional buyers into the new sketch. Overall it helps us understand which factors are the most influential to delivery times. We can also get an idea about performances of managers in different areas, as each area has a corresponding manager.



Some things to note about the sketch above:

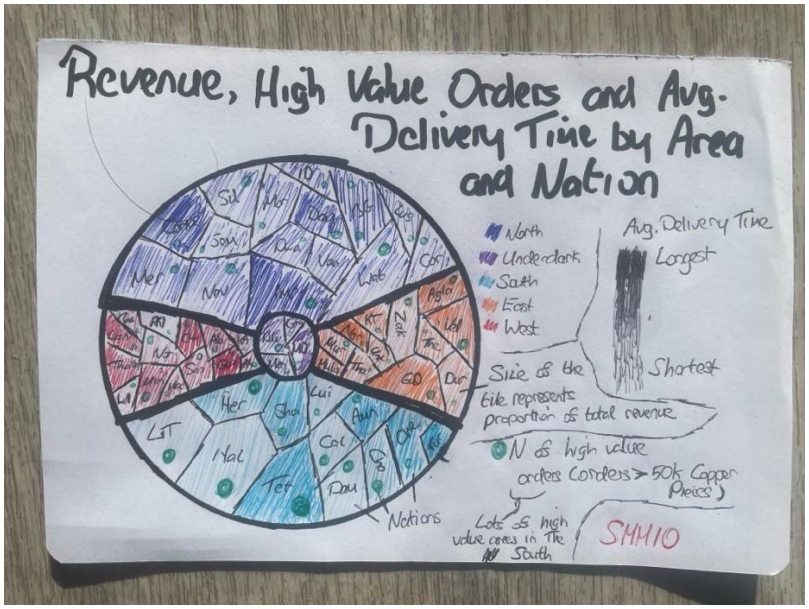
- Delivery time: days till delivery = arrival time – delivery time
- Higher values on the axes mean better performances. So we needed to change delivery time such that higher values on the axes would also mean a better performance (and not a longer delivery time).

Cluster 3: Maps and Geographical Representation



The third main cluster that emerged from combining the sketches of the diverge phase was different kinds of maps. These sketches mainly focused on extracting information about revenue and order value across different locations, categorizing by territories, nations, regions and areas. In SMM1, the circle is split up into five areas representing the five areas in the dataset. These areas are furtherly divided into tiles, with each tile representing a nation. The size of the nations and tiles are proportional to the amount of revenue they take in. Furtherly, each tile has a green dot, with the size of the green dot representing the number orders with a value greater than 50k Copper Pieces coming from that area. We felt this was a very useful piece of information to include, as it helped us to clearly visualize and answer to the question of where high value orders were coming from. VVB8 and VVB9 categorized based on regions and territories and used colours to denote differences in average cart prices and number of orders. Similarly, RK6 shaded each nation with varying degrees of lightness and darkness, depending on the amount of money spent in each area. We decided we'd like to combine these ideas, and that combining all of this information in a map form would be more digestible than other forms. As we denoted the possibility of location playing a key factor in delivery times, we also wanted to incorporate this into the map.

The resulting visualization can be seen below. As before, each tile represents a nation. The different colours reflect the 5 different areas. A green dot for each remains to represent the number of high value orders, and now the lightness of each tile represents the average delivery time of orders in this nation. This graphic delves more into detail about revenue and delivery times across all nations. We can see which nations are gaining our company the most revenue and where our delivery services could be improved. It also gives an idea, whether or not revenue and delivery times correlate. Further interactivity could be added to this map too, when hovering over a tile it could provide a dropdown box of more precise data about the nation.


Revenue, High Value Orders and Avg. Delivery Time by Area and Nation
North
Underclark
South
East
West
Avg. Delivery Time
Longest
Shortest
Size of the tile represents proportion of total revenue
N of high value orders (orders > 50k Copper Pieces)
Lots of high value orders in the South
Nations
SMM10

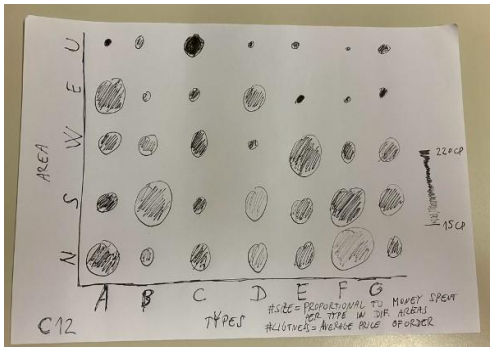
Part 4. Implementation

We met up to select two out of three visualizations, considering the questions we came up in the design phase. The first visualization explores fluctuations in revenue for certain product types and sheds light on average order price per product types and areas, providing insights into product performance and revenue streams. Given the overlap between visualizations 2 and 3, we quickly decided we would only chose one of them. Visualisations 2 and 3 focus on delivery times and revenue respectively per area and per nation and can give an idea whether revenue and delivery times correlate. Visualisation 3 also adds information about the distribution of the high-value orders over nations and gives more detailed information per 57 nation than the 5 areas in visualisation 2. We decided to implement visualizations 1 and 3.

A. Visualization of total revenue and average price of order

A.1 Intended design: Displayed below is a hand-drawn visualization. The size of the circle corresponds to the amount of revenue generated, proportionate to the expenditure on each product type in different area's. It give us the fluctuations in revenue of certain product types per area. We can easily observe which product types are performing exceptionally and which product types are performing poorly in all area's.

Our initial approach involved utilizing the saturation of circles to gain deeper insights into the varying product prices. A darker shade would signify higher product prices in a certain area. However, due to the large number of products (2101 in total), this visualization method became impractical, losing its informativeness. Consequently, we shifted our focus to analyzing product types instead. Now, saturation indicates fluctuations in the average price per order for a specific product type within a particular area, with a darker shade indicating a higher average price per order.

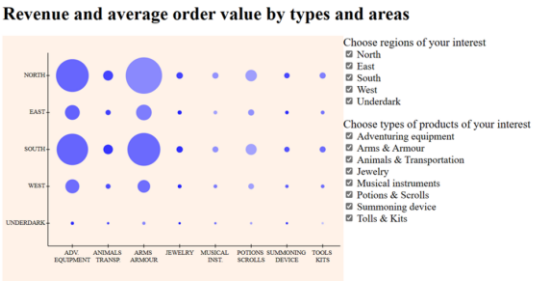


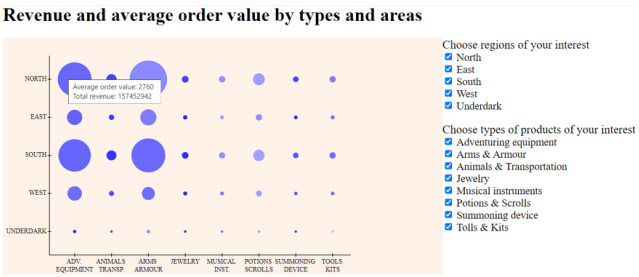
A.2 Actual design

- Visual encoding: We created this visualization using Svelte. Each circle, serving as marks, represents the revenue and average order price of a certain product and area. On the vertical axis, serving as position channel, you'll find five regions (North, South, West, East, and Underdark), while the horizontal axis represents eight product types (Adventuring equipment, Arms & Armour, Animals and transportation, Jewelry, Musical instruments, Potions & Scrolls, Summoning device, and Tools & Kits).

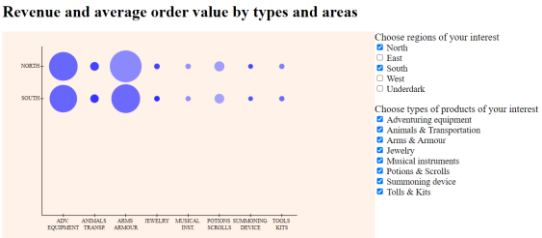
The size of each circle reflects the total revenue generated by a product type in a specific area. Larger circles represent higher revenue. The saturation of blue color within each circle represents the average order value.

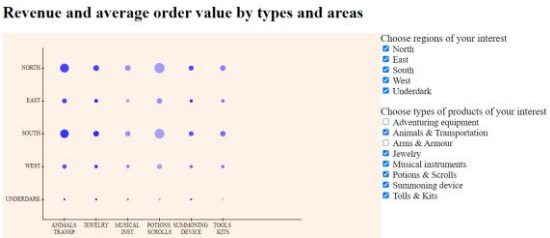
Darker shades of blue indicate higher average price of order. By utilizing these marks and channels effectively, the visualization provides a comprehensive representation of total revenue and average order value across different areas and product types, allowing viewers to quickly interpret and analyze the data.





- Interactions: Several interactive features enhance our visualization. When hovering over a circle with your mouse, a dropdown box gives you more precise data of the total revenue and average order value of that particular area and product type. For more precise comparisons, checkboxes are available on the right side of the visualization. Users can select specific areas and product types to display. The data is than filtered for the subsets of the marked area's and product types. The data circles are grouped together for clearer visualization and comparison.



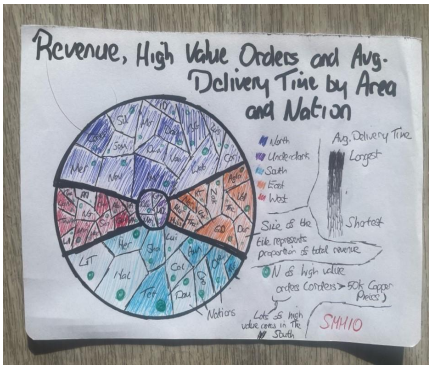


- Intended versus actual design: This implementation serves as a comprehensive realization of the initial idea, capturing the essence and fulfilling the intended functionality as portrayed in the original sketch. Every aspect and detail outlined in the sketch has been translated into this implementation, such that the final product mirrors the envisioned concept laid out in the sketch.

- Link to code for the visualization: [redacted]
https://svelte.dev/repl/fbf46ebde32e410e953adadd1cd5c89d?version=4.2.16

B. Visualization total revenue, delivery time and high-value orders

B.1 The intended design: In the converge phase, we developed a visualization aimed at answering a few key question: does delivery time vary a lot across different locations (nation or area)? Where are our most valuable orders coming from? Do revenue and delivery time correlate?



In this visualization, the 57 tiles correspond to a specific nation, with nations from the same area sharing the same color. The colors represent the five areas, with Underdark positioned at the center and the North, East, West, and South surrounding it, mimicking the orientation of a traditional map.

This visualization offers comprehensive insights into revenue generation across these five areas and 57 nations. The size of each tile corresponds to the revenue generated by the respective nation, with larger tiles indicating higher revenue. The brightness of each tile reflects the average delivery time for orders in that nation, providing valuable information on delivery efficiency. Lighter brightness on the tiles signifies shorter delivery times, indicating higher efficiency compared to darker tiles. Additionally, green dots within the tiles signify the distribution of high-value orders (>50k Copper pieces) across the 57 nations, with larger dots indicating a higher concentration of such orders.

B.2. The actual design

We created this visualization using 3D.

- Visual encoding: In our visualization design, we employed visual encoding techniques to convey key information. Colours were used to denote the areas to which nations belonged. The size of the tiles were utilized corresponding to revenue proportionally.

Delivery time was represented by using the brightness of the tiles. However, the intended approach of looking at average delivery time did not yield significant insights, as there were minimal differences observed between nations. Consequently, we sought a more effective metric and decided to focus solely on delivery times exceeding 20 days. This threshold was selected after analysing the distribution of delivery times, which revealed a majority of deliveries occurring within the 0 to 20-day range, followed by a significant drop. This prompted us to focus on

these outliers. Despite these adjustments, lighter brightness continued to signify better and more efficient delivery times and the darker tiles are the outliers.

Furthermore, we determined high-value orders based on their distribution pattern. High-value orders refer to orders delivered to a nation with a Cart Price exceeding 25,000 Copper Pieces. Larger green dots meant more higher- value orders in that nation.

- Interactions: Hovering over data points reveals tooltips containing information such as the name, total revenue, proportion of late orders, and the number of high- value orders. To facilitate tracking, data points that have been previously checked blur out, distinguishing them from unchecked points. The process resets, when the page is reloaded, with all data points reverting to their original, lighter coloration.

- Intended versus actual design: There was an adjustment in the positioning of the areas driven by considerations of coding simplicity. Defining precise initial localizations (with Underdark in the center) to match the intended design seemed challenging and unnecessary. The repositioning facilitated labelling on the outer edges, contributing to a more visually appealing and refined design. While the layout underwent some modifications, the fundamental principle of proportionality to revenue remained unchanged. The algorithm dictates that all tiles must accurately reflect revenue. Regarding the metric for average delivery time, initial observations revealed no discernible trends. Consequently, we opted to refine the metric to focus exclusively on delivery times exceeding 20 days, providing a clearer indication of performance.

- Link to code: [redacted]

C. Screencast

Link to screencast of the two visualisations: [redacted]

Part 5. Findings

A. Initial patterns and anomalies in the data

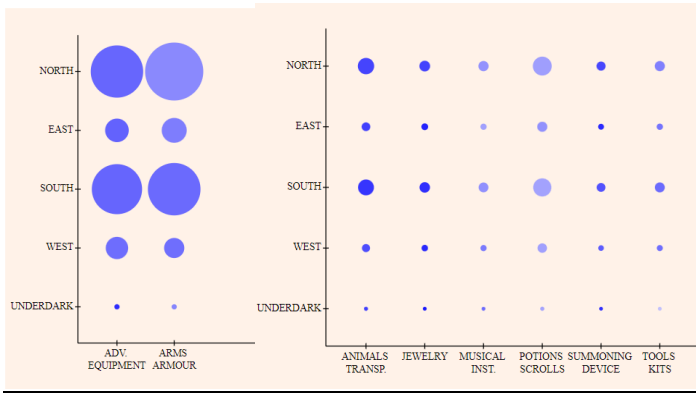
As we examined the data, several patterns emerged. Firstly, we focused on revenue assessment, looking at product categories to identify areas of excellence and those that generated relatively low revenue. Both our visualisations give insight into of the proportion of total revenue by product types, area's or nations.
Secondly, we saw that there was significant variability in delivery times We wanted to better understand the relationship between delivery time, geographical factors (such as areas, regions or nations), product categories and management dimensions. Our visualisation illustrates the relationship between delivery time, geography (area and nation) and product type. Differences in these visualisation will need to be carefully examined because they can vary in terms of customer counts or order quantities, which aren't currently in our visualisations.
Next, product prices showed wide variation across orders. This made us wonder: are these variations influenced by year, geographical factors (area, region, nation), or could there be other reasons behind them. Following the diverge, emerge, and converge phases, no visualization remained to provide a clear answer to this question.
Finally, the analysis showed that order values followed a Pareto distribution, highlighting the significant variance in the value of orders, with some being significantly more valuable than others. Just a small number of orders bring in a big chunk of the total revenue. This idea fits with the saying "the vital few and the trivial many." It's really important to spot and pay attention to these high-value orders because they can help make the most revenue and use their resources wisely.

When analysing the data, some data issues were uncovered in the dataset. Of particular note was the discrepancy between the number of products listed for each order and their corresponding prices. In some cases, there were more products than prices, so the algorithm prioritised assigning values to individual products, ignoring the excess data, or automatically deleted lines with mismatched lengths between product numbers and prices. This results in significant gaps in the dataset for visualisation.
Addressing these issues necessitates data cleaning efforts and collaboration with the company stakeholders to identify anomalies and rectify inaccuracies, ensuring the dataset's suitability for visualization.

B. With the aid of data visualizations, we obtained valuable answers to our questions

<u>Visualization of total revenue and average price of order</u>
Our assessment of revenue involved an exploration of product types to distinguish areas of robust performance from those generating comparatively lower revenue. Certain regions demonstrate notably higher revenue compared to others. Upon comparing total revenue across areas, the visualization clearly illustrates that the North and South yield the highest revenue, followed by the East and West. Unsurprisingly, Underdark, being the smallest area, records the lowest revenue. Hence, caution is advised in drawing conclusions, as the size of the area influences revenue potential due to the number of potential customers.

Further analysis reveals substantial disparities in revenue generation among product types. While some, like 'Adventuring equipment' and 'Arms and Armour', contribute significantly to revenue, others such as 'Animals and Transportation' or 'Potions and Scrolls' exhibit considerably lower returns. Product types like 'Tools and Kits' and 'Summoning devices' consistently generate minimal revenue across all areas. The company should consider reallocating resources and logistics efforts towards high-performing product types, while phasing out those with marginal contributions to revenue to optimize operational efficiency.



Another question we asked ourself was around the fluctuations in product prices. Initially, we aimed to compare prices across all areas, regions, and nations. Due to constraints in our initial sketch, accommodating all 2101 products proved unfeasible. Consequently, we shifted our focus to analysing product types and their average order prices instead. The varying shades in the visualization signify differences in average order prices among product types. While both 'Adventuring Equipment' and 'Arms and Armour' contribute similarly to revenue in the North area, they exhibit slight disparities in average order price. 'Adventuring Equipment', indicated by a darker shade of blue, reflects a higher average order price compared to 'Arms and Armour'. Customers who buy 'Adventuring Equipment' tend to spend more and often buy additional items or in bulk, unlike those who buy 'Arms and Armour'. This could be influenced by differences between area's in targeted marketing or discounts encouraging supplementary purchases. It's also interesting to examine product types with low revenue and their disparities in average order prices. We note that customers buying 'Musical Instruments' and 'Potions and Scrolls' tend to have lower average order prices than for example Jewellery. Jewellery is purchased infrequently (small revenue) despite its high average price, indicating its niche appeal. The average price per order can give an idea about popular product types, these are the products that are more expensive or purchased in larger quantities per order. For a comprehensive analysis, it is crucial to also look at individual product prices which are not in our visualisation. Those will reveal whether customers prefer to order large quantities of one type of item (e.g. 200 potions) or opt for smaller quantities but more frequent orders.

Visualization total revenue, delivery time and high-value orders

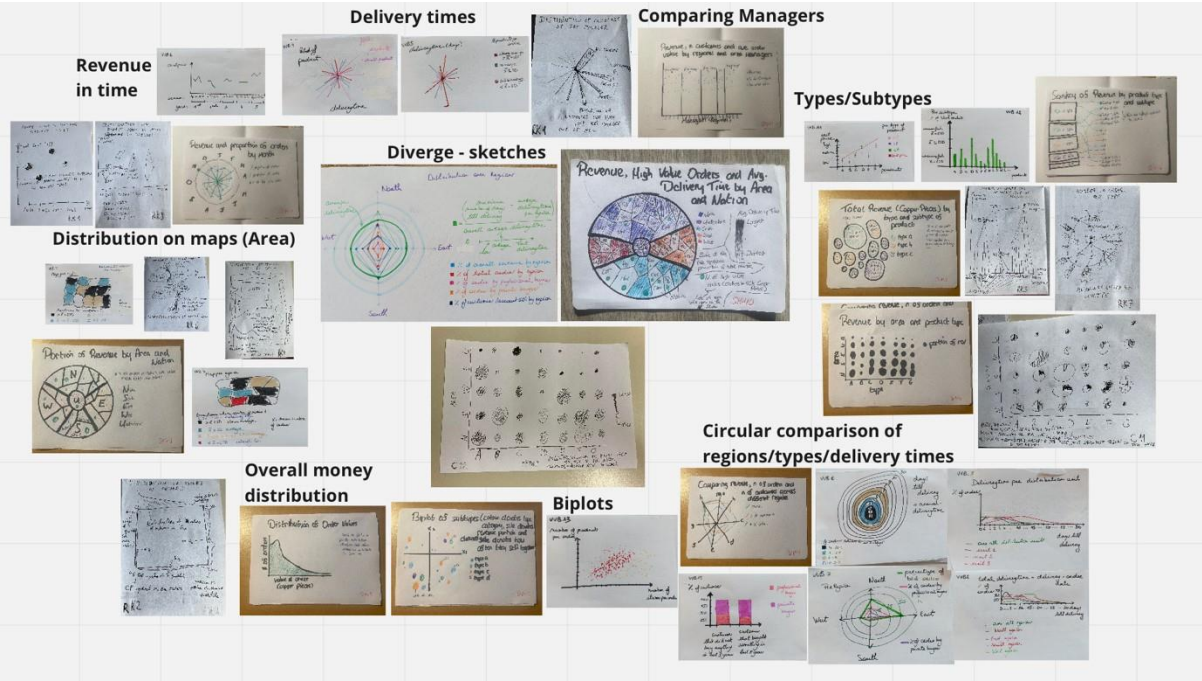
This visualization serves as a tool for identifying revenue-generating opportunities and potential for enhancing delivery services to better meet customer needs and expectations. The varying tile sizes for the five areas and 57 nations provide insights into each region and nation's performance relative to total revenue. The North area stands out as the highest revenue-generating area, with Amn in the South being the largest nation tile making a significant contribution compared to other nations in the area.

This visualization aids in pinpointing area's and nations requiring improved delivery services. Darker tiles indicate a higher proportion of delayed delivery times (over 20 days), particularly noticeable in the East area. Investigating this region is crucial to understand underlying issues. Conducting intra-area analysis is crucial for discerning performance discrepancies, where some nations excel while others lag behind within a given area. High total revenue doesn't necessarily correlate with efficient delivery performance for a nation. Take Silveyrmoon and Sembia, for instance; despite their significant revenue, they face longer delivery times, indicating a requirement for optimization or investment in logistics within these nations. Leveraging these insight can enhance the performance of underperforming area's and nations to meet or exceed standards.

We also wanted to identify the origins of our most valuable orders. Many nations display small green dots, some barely visible, while others have significantly larger ones. A handful of nations, regardless of their size, are responsible for nearly all high-value orders, highlighting how a small fraction of orders can greatly impact total revenue. For instance, 'Amn' stands out as a hub for high-value orders. By targeting these key nations, this company can strategically allocate resources and optimize revenue generation.

Part 6. Individual Contributions

During the design phase, our team convened several times, both online and in person, to discuss and develop visualizations for the three clusters. Each team member took responsibility for crafting a visualization for one of the clusters. [redacted] initiated the report, detailing the process undertaken for cluster 2. [redacted] and [redacted] then contributed their sections on clusters 1 and 3, respectively.

In the implementation phase, we individually reviewed the assignment before our initial meeting. During this meeting, we divided the tasks among ourselves [redacted] xpressed interest in refining visualization 3, whil [redacted] opted to further develop visualization 1. [redacted] as responsible for compiling the final report and providing coding support. During our subsequent meeting, which lasted several hours, we collaborated on resolving coding challenges. We set a deadline for completing the visualizations and recording screencasts. As planned, [redacted] finalized the report and sought feedback from [redacted] before submitting the assignment.

Appendix A. Sketches overview